path 1

1) (10.71, -7.75 , 5.2)

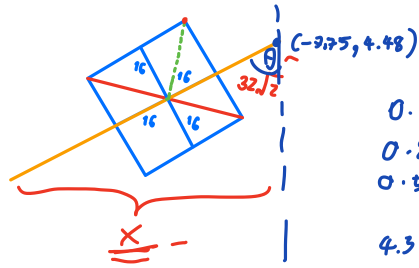
2) (10.71 , -9.7 , 5.2)

3) (11.2746 , -10.04 , 5.44)   Q (0, 0, -0.707, 0.707)

4) (11.27460, -9.42284, 4.48)

5) (11.27460, -7.84178, 4.48)

6) (11.27460, -7.84178, 4.96538)



0.41
0.26
0.58

4.32
4.48
0.16

(10.71, -7.75, 4.48)

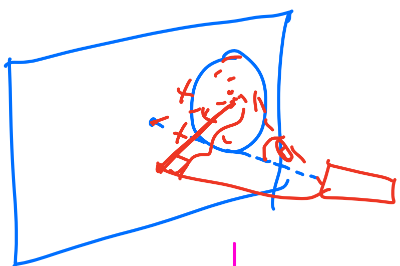
1) (11.39, -9.39, 4.48)

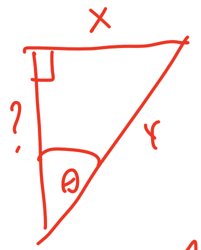
0.0422   2) (11.2746, -10.04, 5.41) Q (0, 0, -0.707, 0.707) target
11.3168

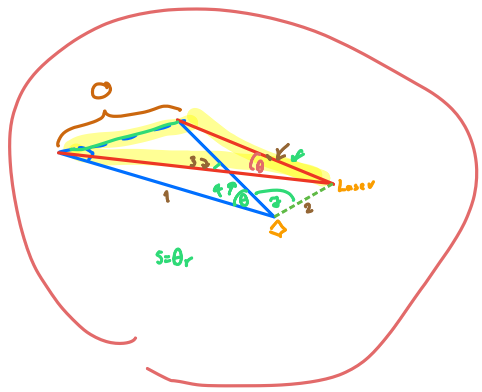
3) (10.71, -9.7, 5.2)

4) (10.71, -7.75, 5.2)

5) (11.27460, -7.84178, 4.96538) Q (0, 0, -0.707, 0.707)

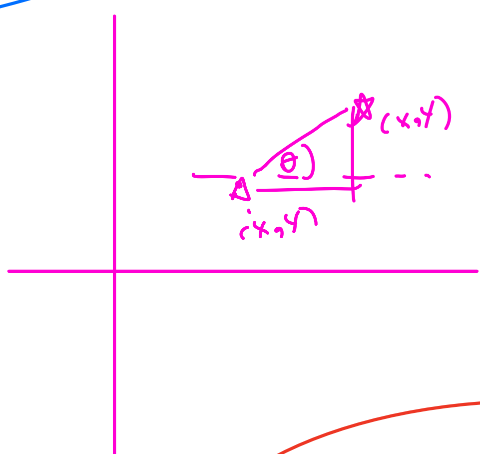


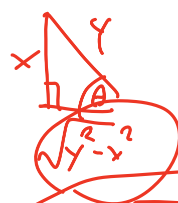




$$\frac{1}{\cancel{\sin 90}\,y} = \frac{\sin\theta}{x}$$

$$\boxed{\sin\theta = \frac{x}{y}}$$





$$\cos\theta = \frac{\sqrt{y^2 - x^2}}{y}$$

ดูละจ้า

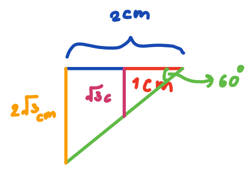
1) ระยะจากศูนย์กลางถึงกล้อง

2) ระยะจากศูนย์กลาง – ศูนย์กลางวงกลม

ถ้าทำเป็นพิกัดได้ ก็เอา
ถ้าไม่ได้ของเป็น ท
ทได้ก็ง่ายดี

Name	Explanation
HazCam	A monochrome camera for detecting obstacles within 30 cm
NavCam	A monochrome camera for image data processing and taking a photo after sending finish command
SciCam	A color camera for taking a video (The participants cannot utilize this. In the final round SciCam is used to take color videos and acts as Astrobee's eye.)
DockCam	A monochrome camera for docking to the docking station (In the simulator, it takes videos of the rear.)
PerchCam	A monochrome camera for grabbing a handrail (You can create a program with this camera, if needed.)
Laser Pointer	Irradiating the target (It has a distance from NavCam. Be careful when you create a program of image data processing. For details of distance, please refer to Programming Manual clause 5.4)
Flashlight	Use this when reading QR code
Speaker	Report "mission complete" to an astronaut by your voice

Table 3.6.1-1 Rendering of Astrobee



$\tan 60° = \frac{x}{2}$

$x = 2\sqrt{3}$

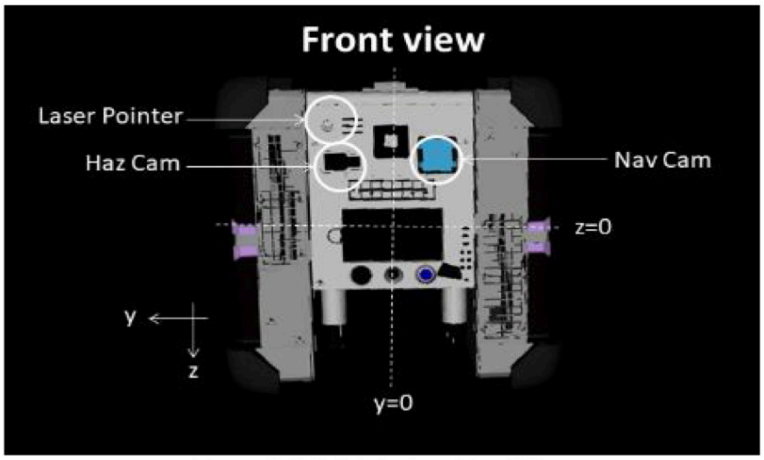
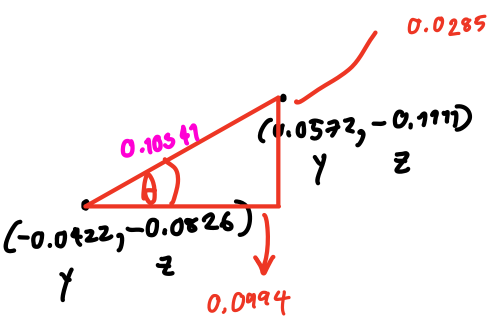


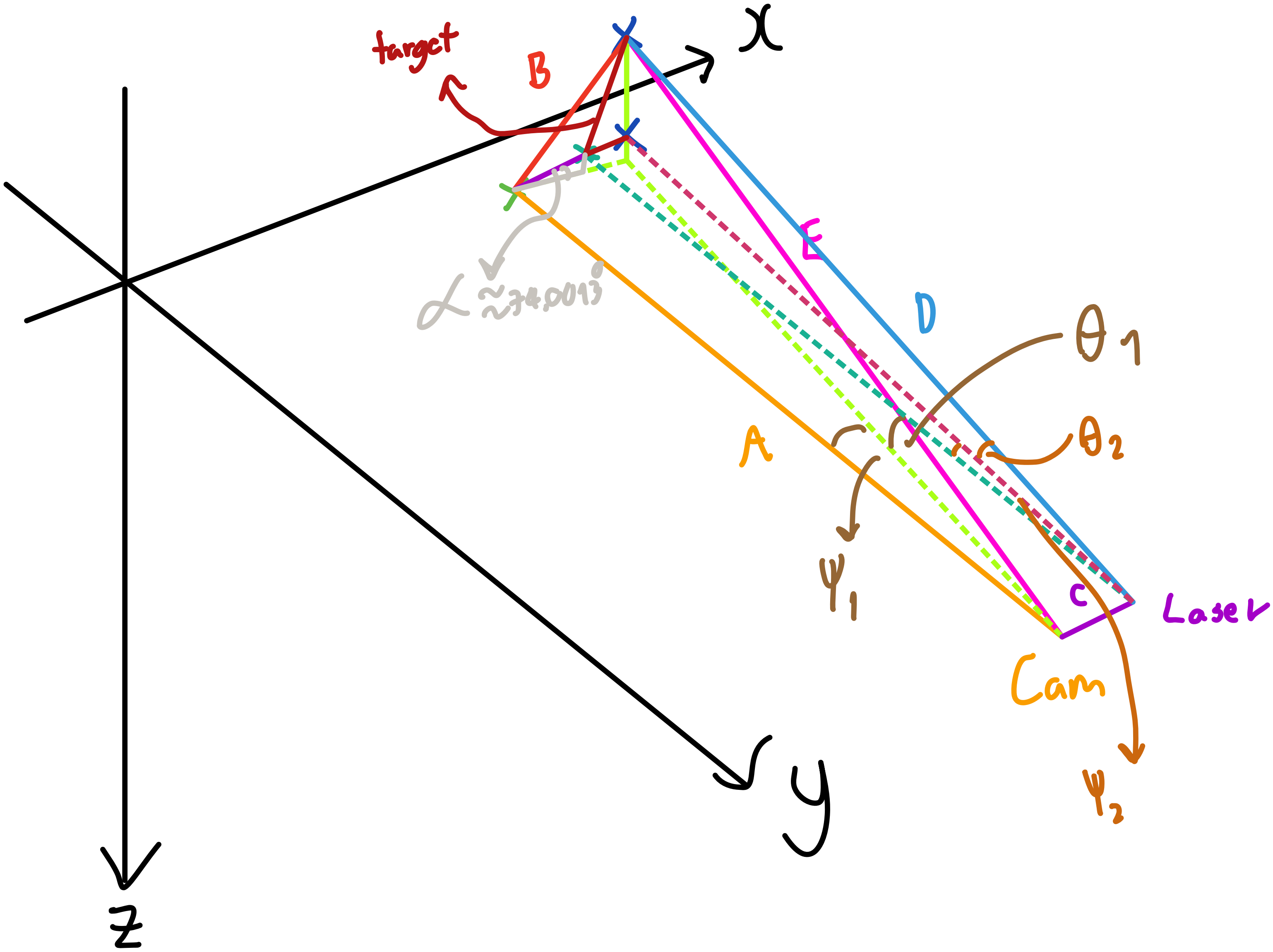
Figure 3.6.1-2 Astrobee Front View

Table 3.6.1-1 Distances from center point

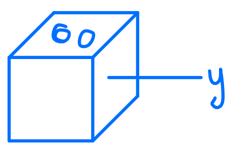
	x[m]	y[m]	z[m]
Nav Cam	0.1177	-0.0422	-0.0826
Haz Cam	0.1328	0.0362	-0.0826
Laser Pointer	0.1302	0.0572	-0.1111



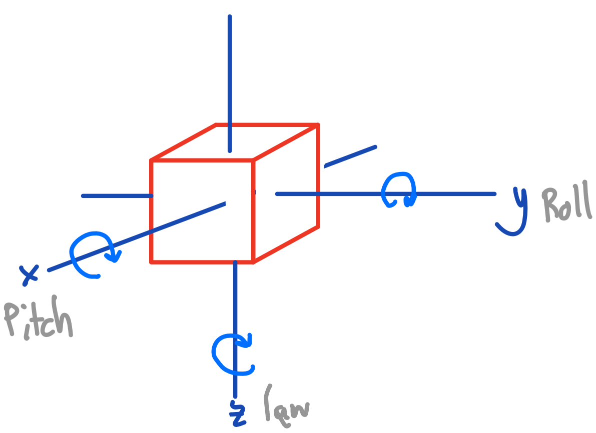
$\theta \approx 74.0073°$



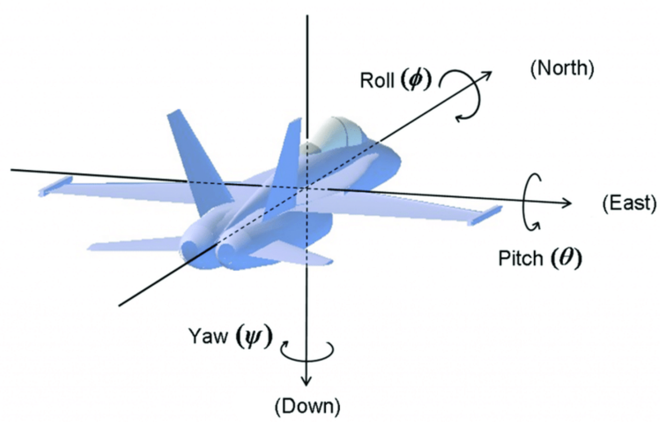
x
target
B
E
D
θ1
θ2
A
ψ1
ψ2
C
Laser
Cam
y
z



y



y Roll
x
Pitch
z Yaw



Roll (φ)
(North)
(East)
Pitch (θ)
Yaw (ψ)
(Down)

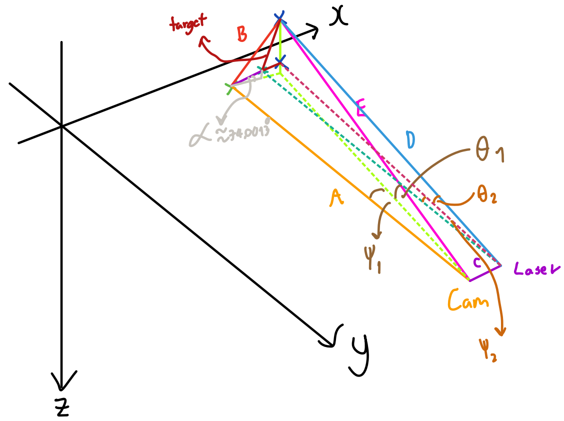
A. ระยะ Cam ถึงศูนย์กลางภาพ (จาก image detect)

B. ระยะจากศูนย์กลางภาพถึงศูนย์กลางวงกลม (จาก image detect)

C. ระยะจากกล้องถึง Laser (Constant)

D. ระยะเลเซอร์ถึงศูนย์กลางวงกลม (Calculate)

E. ระยะจาก Cam ถึงศูนย์กลางวงกลม (Calculate)



Path1

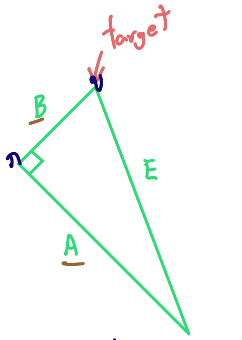
Solution

ศูนย์กลางกล้อง = ก

target = ข

3 step

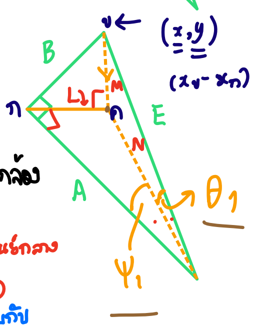
1: Cam to target



① $E^2 = A^2 + B^2$

$E = \sqrt{A^2 + B^2}$

(เอา E ใช้ต่อ)



② เปลี่ยนค่าแกน z ของวงกลมให้เท่ากันศูนย์กลางกล้อง

2.1) หาระยะห่างระหว่างศูนย์กลางกล้องถึงจุดนั้น (L)

เอาพิกัดแกน x มาลบกัน

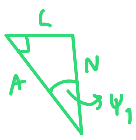
2.2) หาระยะจากจุดนั้นถึงกล้อง (N)

$N^2 = L^2 + A^2$

$N = \sqrt{L^2 + A^2}$

(เอา N ใช้ต่อ)

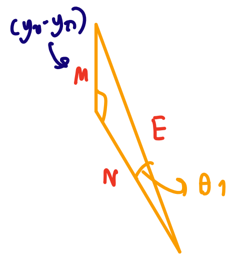
2.3) หา ψ_1



$\frac{\cancel{\sin 90^\circ}\ 1}{N} = \frac{\sin \psi_1}{L}$

$\sin \psi_1 = \frac{L}{N}$

$\therefore \psi_1 = \sin^{-1}\left(\frac{L}{N}\right)$

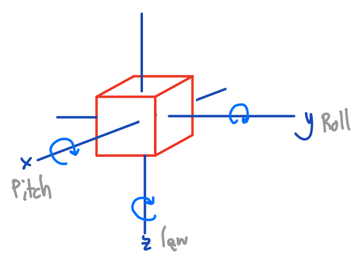


2.4) หาระยะ z จากการลบพิกัดแกน z ของศูนย์กลางวงกลมกับศูนย์กลางกล้อง

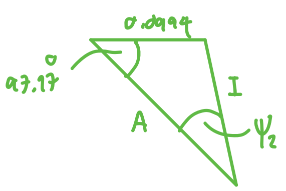
$\frac{\sin 90^\circ}{E} = \frac{\sin \theta_1}{Z}$

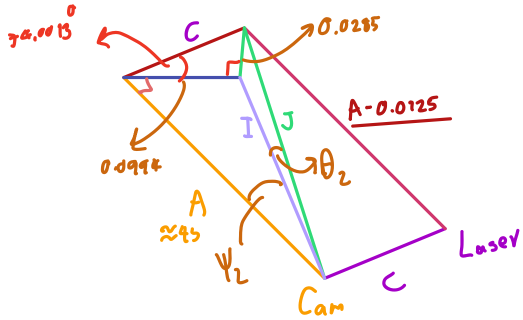
$\sin \theta_1 = \frac{M}{E}$

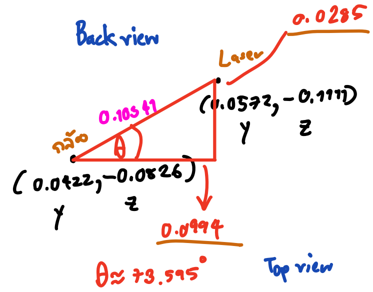
$\therefore \theta_1 = \sin^{-1}\left(\frac{M}{E}\right)$



2: Cam to Laser







① ลดพิกัดแกน z ของ Laser ให้เท่ากับ Cam
(0.0285)

② หาระยะห่างระหว่างศูนย์กลางของ Cam ถึง Laser (0.0994)

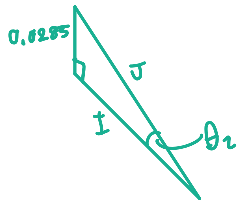
③ หา I

$$I = \sqrt{A^2 + 0.0994^2}$$

④ หา ψ_2

$$\frac{\sin 90^\circ}{I} = \frac{\sin \psi_2}{0.0994}$$

$$\psi_2 = \sin^{-1}\left(\frac{0.0994}{I}\right)$$



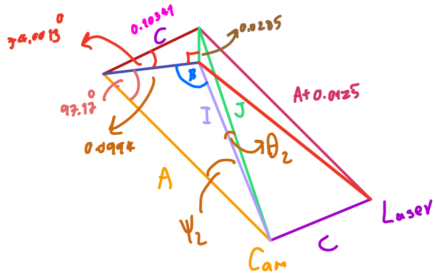
⑤ หา J

$$J = \sqrt{I^2 + 0.0285^2}$$

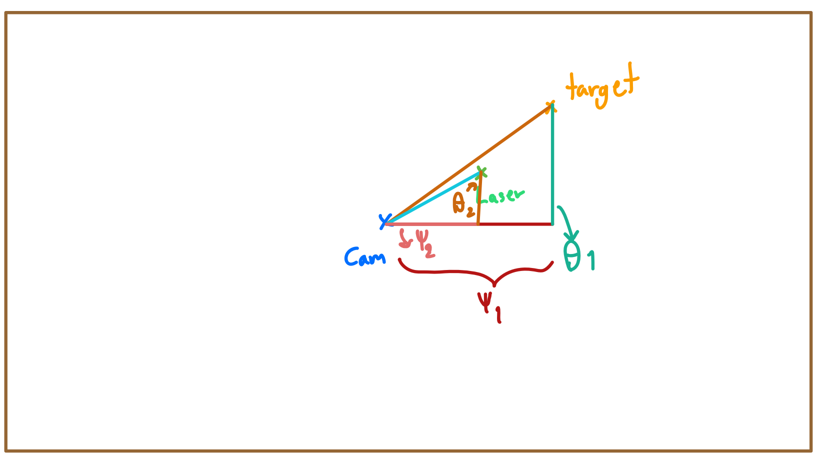
⑥ หา θ_2

$$\sin\theta_2 = \frac{0.0285}{J}$$

$$\theta_2 = \sin^{-1}\left(\frac{0.0285}{J}\right)$$



③ Total Rotation

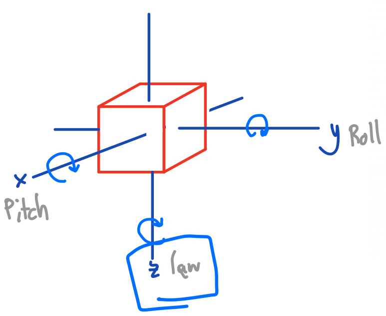


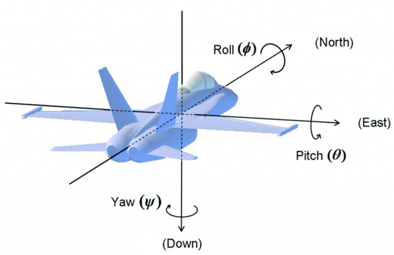
$\boxed{\psi_L} = \psi_1 - \psi_2$

$\boxed{\theta_L} = \theta_1 - \theta_2$

ใช้

แปลงเป็น Quaternion





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Flashlight	Use this when reading QR code
Speaker	Report "mission complete" to an astronaut by your voice

Table 3.6.1-1 Rendering of Astrobee

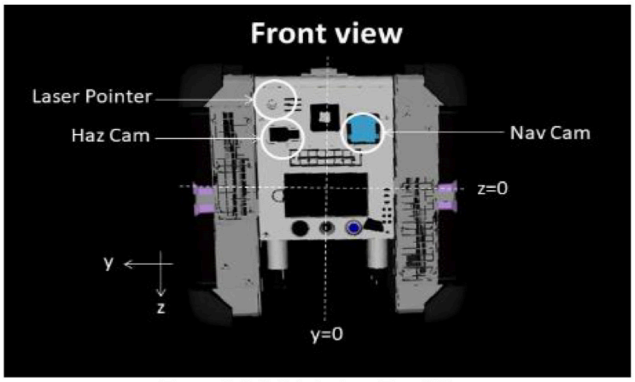


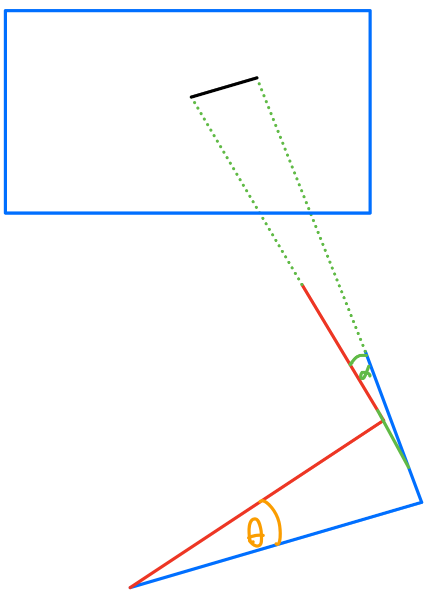
Figure 3.6.1-2 Astrobee Front View

$\sqrt{0.0572^2 + 0.1302^2}$

$= 0.1422106884$

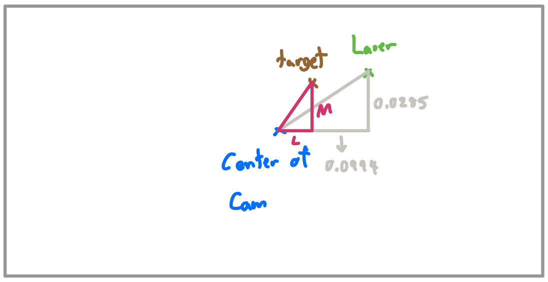
Table 3.6.1-1 Distances from center point

	x[m]	y[m]	z[m]
Nav Cam	0.1177	-0.0422	-0.0826
Haz Cam	0.1328	0.0362	-0.0826
Laser Pointer	0.1302	0.0572	-0.1111

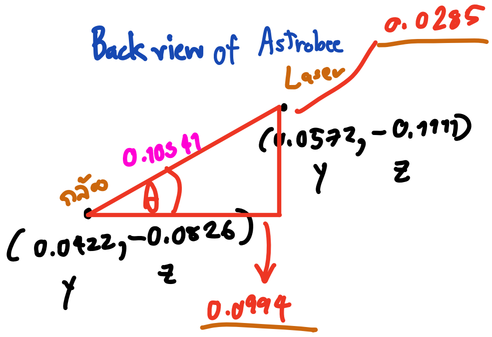


Path2

Front View of Map

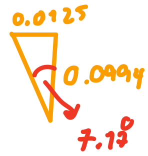


Back view of Astrobee

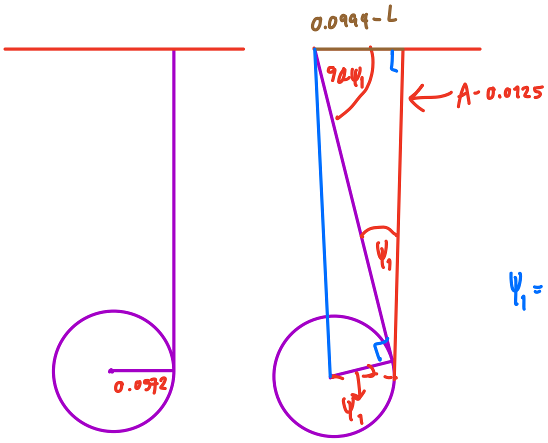


$\theta \approx 73.595^\circ$

Top view



Step 1 Yaw (Rotation):



$$\psi_1 = \tan^{-1}\left(\frac{0.0994 - L}{A - 0.1177}\right)$$

Step2 Pitch (Rotation)

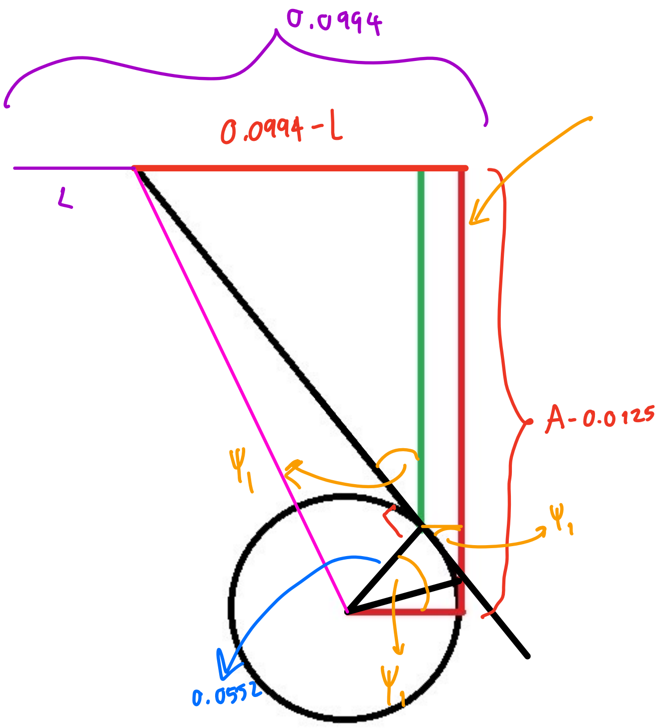
$$\theta = \tan^{-1}\left(\frac{0.0285 - M}{A - 0.0125}\right)$$

Step3 : Total Rotation

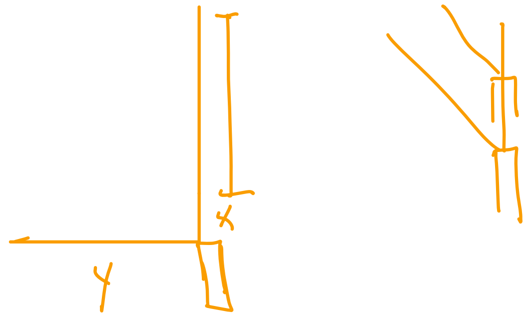
$$\psi = (270^\circ \pm \psi_1)$$

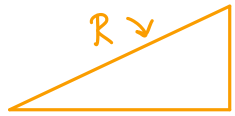
→ ถ้า target อยู่ซ้าย Laser ψ_1 ต้องติดลบ

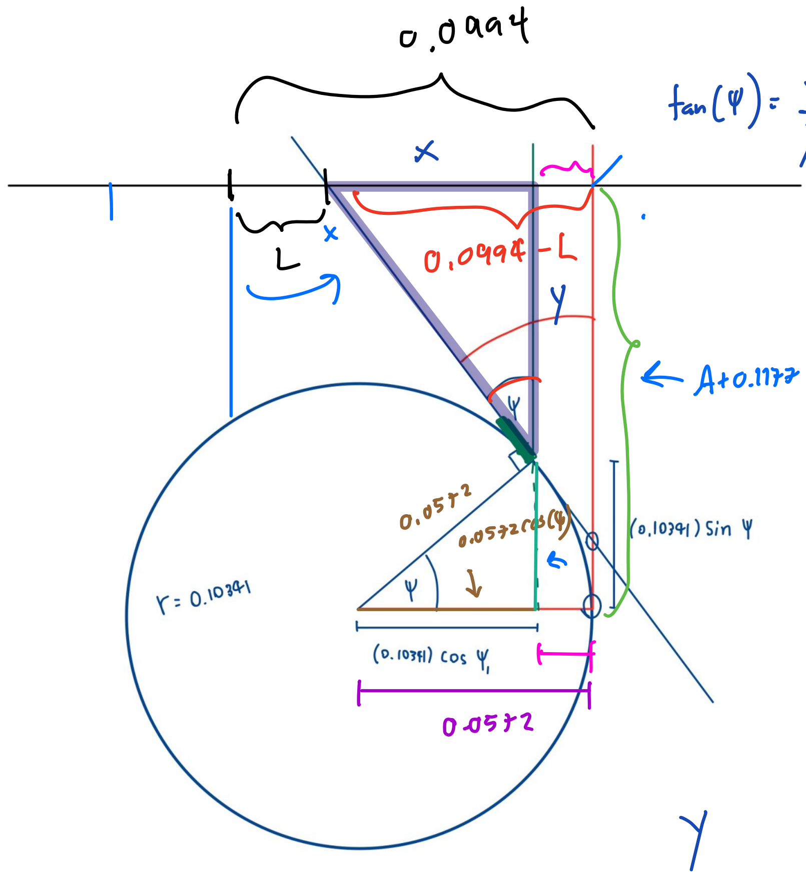
$$\theta = \tan^{-1}\left(\frac{0.0285 - M}{A - 0.0125}\right)$$



$$\tan(\psi_1) = \frac{0.0994 - L - (0.0562 - 0.0552\cos\psi_1)}{A + 0.0125 - \sin(\psi_1)(0.0552)}$$

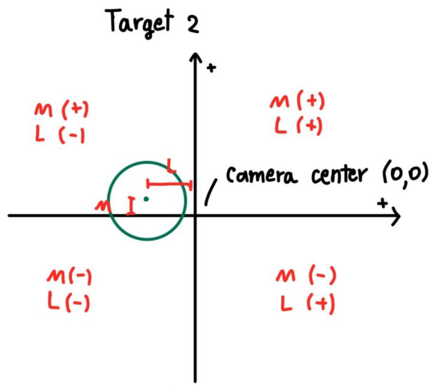






$$\tan(\psi) = \frac{x}{y}$$

$$x = 0.0994 - L - (0.0572 \cdot 0.0572\cos(\psi))$$



A ระยะจากกล้อง → target

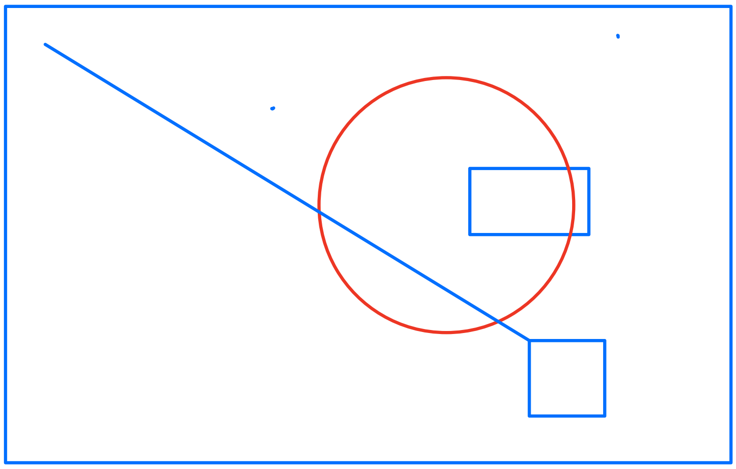
$$\tan(yaw) = -\frac{(0.0994 - L) - (0.0572 - \cos(yaw)\cdot 0.0572)}{(A + 0.1177) + \sin(yaw)\cdot 0.0572}$$

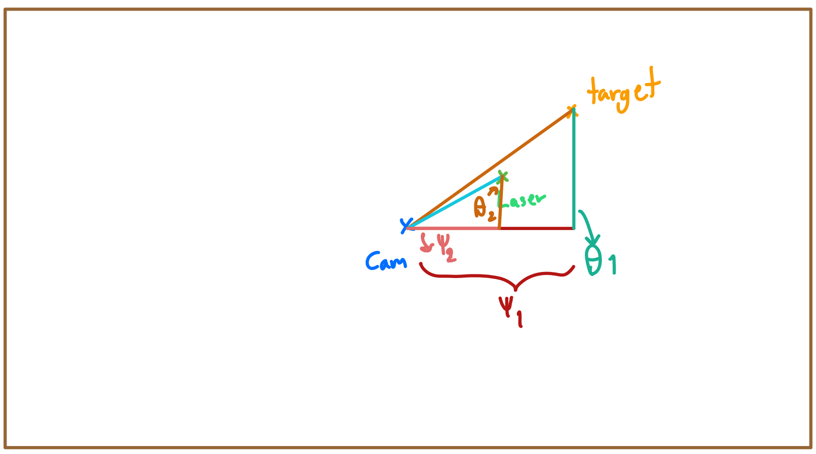
เครื่องหมายเลยเปลี่ยน

$$\tan(pitch) = -\frac{(0.0285 - M) - (0.1111 - \cos(pitch)\cdot 0.1111)}{(A + 0.1177) + \sin(pitch)\cdot 0.1111}$$

$$\tan(\psi) = \frac{(0.0994 - L) - (0.0572 - \cos(\psi)\cdot 0.0572)}{(A + 0.1177) - (\sin(\psi_1) 0.0572)}$$

Yaw ติดลบ





A. ระยะ Cam ถึงศูนย์กลางภาพ

B. ระยะจากศูนย์กลางภาพถึงศูนย์กลางวงกลม

C. ระยะจากกล้องถึง Laser

D. ระยะเลเซอร์ถึงศูนย์กลางวงกลม

E. ระยะจาก Cam ถึงศูนย์กลางวงกลม

Path 1

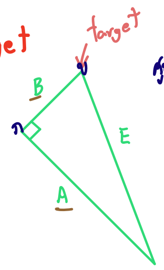
Solation

3 step

1: Cam to target

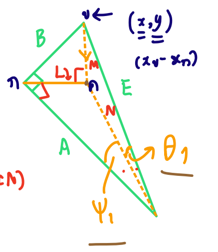
① $E^2 = A^2 + B^2$

$E = \sqrt{A^2 + B^2}$
(เอา E ไปใช้ต่อ)



ศูนย์กลางกล้อง = ก
target = ข

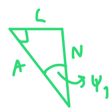
② เปลี่ยนค่าแกน y ของจุดกลมให้เท่ากับศูนย์กลางกล้อง



2.1) หาระยะห่างระหว่างศูนย์กลางกล้องถึงจุดนั้น (L)
เอาพิกัดแกน x ลบกัน

2.2) หาระยะจากจุดนั้นถึงกล้อง (N)

$N^2 = L^2 + A^2$
$N = \sqrt{L^2 + A^2}$
(เอา N ไปใช้ต่อ)

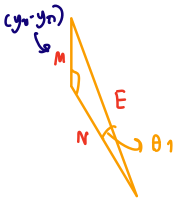


2.3) หา ψ_1

$\frac{\cancel{\sin 90}^{\,1}}{N} = \frac{\sin\psi_1}{}$

$\sin\psi_1 = \frac{L}{N}$

$\therefore \quad \psi_1 = \sin^{-1}\left(\frac{L}{N}\right)$

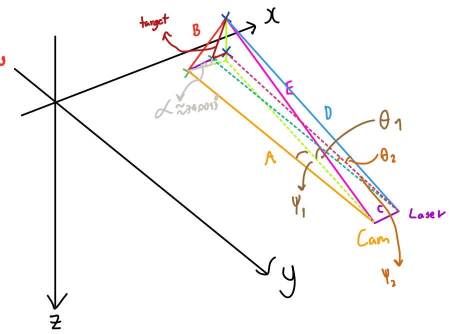


2.4) หาระยะ z จากการลบพิกัดแกน z ของศูนย์กลางวงกลมกับศูนย์กลางกล้อง

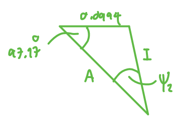
$\frac{\sin 90^\circ}{E} = \frac{\sin\theta_1}{Z}$

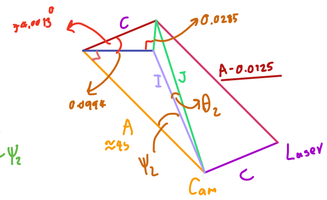
$\sin\theta_1 = \frac{M}{E}$

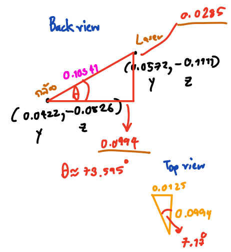
$\therefore \theta_1 = \sin^{-1}\left(\frac{M}{E}\right)$



2: Cam to Laser







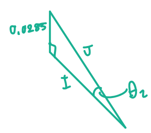
① ลดพิกัดแกน z ของ Laser ให้เท่ากับ Cam
(0.0285)

② หาระยะห่างระหว่างศูนย์กลางของ Cam ถึง Laser (0.0994)

③ หา I
$I = \sqrt{A^2 + 0.0994^2}$

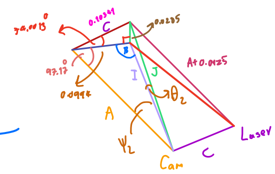
④ หา ψ_2
$\frac{\sin 90^\circ}{I} = \frac{\sin\psi_2}{0.0994}$

$\psi_2 = \sin^{-1}\left(\frac{0.0994}{I}\right)$



⑤ หา J
$J = \sqrt{I^2 + 0.0285^2}$

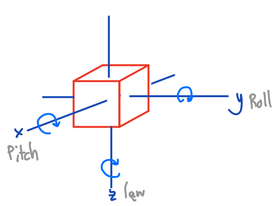
⑥ หา θ_2
$\sin\theta_2 = \frac{0.0285}{J}$
$\theta_2 = \sin^{-1}\left(\frac{0.0285}{J}\right)$



3 Total Rotation

$\boxed{\begin{matrix}\psi_L = \psi_1 - \psi_2 \\ \theta_L = \theta_1 - \theta_2\end{matrix}}$

ใช้แปลงเป็น Quaternion



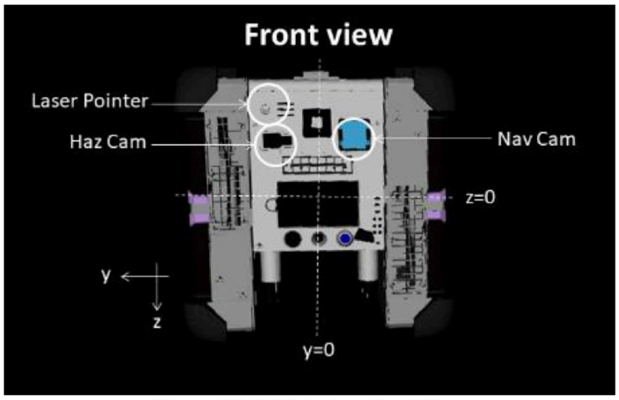
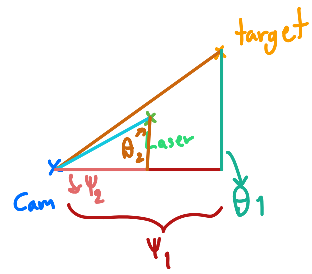
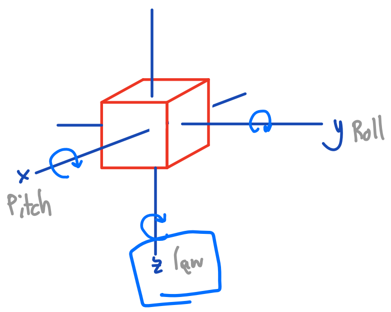


Figure 3.6.1-2 Astrobee Front View

Table 3.6.1-1 Distances from center point

	x[m]	y[m]	z[m]
Nav Cam	0.1177	-0.0422	-0.0826
Haz Cam	0.1328	0.0362	-0.0826
Laser Pointer	0.1302	0.0572	-0.1111





Solution 3 step

1: Cam to target

2: Cam to Laser

3: Total Rotation